Alhoussine Khallil

Ottawa, ON, Canada | housinekhallil@gmail.com | +1 (819) 319 7317
https://www.linkedin.com/in/alhoussine-khallil | https://github.com/alhoussine04

Computer Science graduate with hands-on experience building, testing, and optimizing system-level and AI-enabled software. Skilled in Linux, Python, C/C++, JavaScript, multithreading, and networked applications. Passionate about designing reliable, scalable, and secure software systems, debugging under constraints, and delivering tools that shape system behavior.

Technical Skills

Networking & Systems: Linux (Debian), TCP/IP fundamentals, DNS, process/network control, system-level debugging
Languages: Python, C, C++, JavaScript, SQL
Tools & Platforms: Git, Bash, GCC, Qt, Electron, Raspberry Pi, QNX RTOS
Testing & Automation: Integration testing, performance benchmarking, multithreading, system validation
Libraries & Databases: TensorFlow, OpenCV, NumPy, PostgreSQL

Projects & Software Development Experience

Honours Project (A+) — Real-Time Emergency Sound Detection System | Python, TensorFlow Lite, Raspberry Pi, Linux | [Github] [Paper] | August 2025

Designed and deployed a real-time, Linux-based system that pushed performance limits under CPU and memory constraints, emphasizing reliability, observability, and testability.

- Built a real-time, edge-based audio classification system to detect critical emergency sounds with 86.7% accuracy.
- Designed modular architecture separating data capture, preprocessing, inference, and alerting.
- Developed multithreaded alerting and benchmarking frameworks for CPU and memory-constrained environments.
- Created cross-platform test scripts for hardware integration validation.

RaDoTech Health Monitoring System | C/C++, Qt, PostgreSQL | [Github] | December 2024

Worked in a 4-member team on a large C/C++ project emphasizing design correctness and system reliability.

- Developed a cross-platform simulator using Qt and C/C++, contributing over 1,500 lines of production-quality code.
- Translated functional requirements into UML use case, sequence, and state diagrams prior to implementation.
- Collaborated cross-functionally to translate system requirements into UML-based designs and tested C++ implementations.
- Worked closely with teammates to review designs, debug system-level issues, and iteratively improve software quality.

LabCoat Workspace Manager | Electron, Node.js, NextDNS | [Github] | February 2026

Built an Electron desktop app that enforces workspace rules for blocked, protected, auto-launch, and frozen processes with real-time monitoring.

- Implemented program discovery and thumbnail visibility to streamline configuration and app selection.
- Added network controls via hosts-file blocklists and NextDNS category filtering with a local DNS proxy and system DNS switching.
- Designed IPC-driven architecture separating UI, enforcement, and background discovery with persistent user-data storage.

Education

Bachelor of Computer Science (Honours)
Carleton University — Ottawa, ON | Summer 2025

Achievements

Carleton University Entrance Scholarship
Award of Excellence for International Students